

MATH 3042- Quiz 9- Solutions

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Time: 30 minutes

Full Name:

Set:

Question 1

A networking team investigates whether distance between servers (km) affects network latency (ms).

distance = c(50,100,150,200,250,300,350,400)

latency = c(5,9,14,18,24,29,35,40)

Which graphs, test outputs, or considerations are needed to check on all the requirements of correlation test?

To validate a **correlation** test in R, we need a Scatterplot with distance on the x-axis and latency on the y-axis. We would look for presence of linear trend and lack of outliers.

For **normality** check we perform `shapiro.test(distance)` and `shapiro.test(latency)` and hope for a p-value greater than 5% in each test.

R Output:

cor = 0.998
p-value < 0.0001

$r = 0.998$ (strong correlation)

$H_0: \rho = 0$

$H_A: \rho \neq 0$

$$t = \frac{0.998 - 0}{\sqrt{\frac{(1 - 0.998^2)}{8 - 2}}} = 38.7$$

Tasks:

- Describe the relationship (direction, strength, linearity)
- Write the H_0 and H_A for a correlation test
- Calculate the test statistic
- Write the filled-in R function needed to calculate the p-value
- Use the given p-value to make a decision
- A manager concludes: "Distance directly causes latency." Explain whether this conclusion is fully justified

$$p\text{-value} = 2 \left(\text{pt}(-38.7), 8 \right)$$

Decision: Reject H_0

Correlation is not to be confused with causation. We can say "There is a strong association between distance and latency." but we cannot claim that distance causes latency. Note that even though distance may cause latency, but our `cor.test()` is not suitable for making conclusions about causality.

Question 2

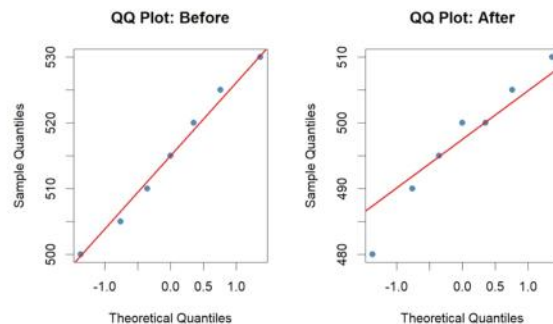
A system upgrade is suspected to reduce memory usage (MB):

before <- c(500, 520, 510, 530, 505, 515, 525) $d = \text{before} - \text{after}$
 after <- c(480, 500, 495, 510, 490, 500, 505)

Tasks:

- State hypotheses
- Check normality using R output
- Should we perform paired t-test or independent t-test?
- Manually perform the test
- Conclusion at $\alpha = 0.05$

$d = \text{before-after}$
 summary(d)
 Min. 1st Qu. Median Mean 3rd Qu. Max.
 15.00 15.00 20.00 17.86 20.00 20.00
 sd(d) = 2.672612



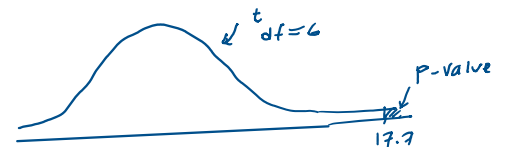
$$\begin{aligned} a) \quad H_0: \mu_d &= 0 \\ H_A: \mu_d &> 0 \end{aligned}$$

Right-tailed test

b) both QQ plots seem reasonably normal

c) paired t-test

$$d) \quad t = \frac{\bar{d} - \mu_d}{\frac{sd}{\sqrt{n}}} = \frac{17.86 - 0}{\frac{2.67}{\sqrt{7}}} = 17.7$$



$$p\text{-value} \approx \phi$$

Decision: we reject H_0

Conclusion: The upgrade significantly reduces memory usage.

Question 3

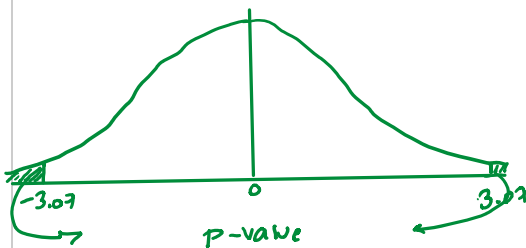
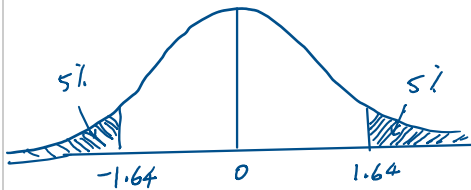
- System A: 75/150 users enabled 2FA
- System B: 120/180 users enabled 2FA

Is the difference in proportion? Use $\alpha = 10\%$

Tasks:

- State the Hypotheses
- Check normal approximation conditions
- Manually calculate the test statistic.
- State the p-value as a probability statement and draw it on a graph.
- State your decision and conclusion

> `qnorm(0.90)` = 1.281552
 > `qnorm(0.95)` = 1.644854
 > `qnorm(0.975)` = 1.959964



normality check

Is $(150)(0.5) \geq 10$? Yes

Is $(180)(0.67) \geq 10$? Yes

Is $(180)(0.33) \geq 10$? Yes

$$\hat{p}_A = 0.5$$

$$\hat{p}_B = 0.67$$

$$H_0: P_A - P_B = 0$$

two tailed test

$$H_A: P_A - P_B \neq 0$$

$$\hat{p} = \frac{75 + 120}{150 + 180} = 0.59$$

$$\hat{q} = 1 - \hat{p} = 0.41$$

$$Z_{\text{test}} = \frac{(0.5 - 0.67) - 0}{\sqrt{(0.59 \times 0.41) \times (\frac{1}{150} + \frac{1}{180})}} = -3.07$$

The p-value is less than $\alpha = 10\%$.

we reject H_0

$$\begin{aligned} \text{p-value} &= \Pr(|Z| \geq 3.07) \\ &= 2 \left(\text{pnorm}(-3.07) \right) \end{aligned}$$

Decision: we reject H_0

Conclusion:

There is a **significant difference** in proportions

- System B has higher adoption ($\approx 66.7\%$ vs 50%)
- System B performs better for 2FA enablement

Two populations:

Formula to test the Difference in Two Dependent (Paired) Populations:

$$t_{test} = \frac{\bar{d} - \mu_d}{\left(\frac{s_d}{\sqrt{n}}\right)} \text{ degree of freedom } \nu = n - 1$$

Correlation test

$$\text{Correlation Test statistic: } \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}, \text{ use } t - \text{distribution, } df = n - 2$$

t test statistic (Test the Difference in Means when the variances are different)

$$t_{test} = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

where the degree of freedom is approximately equal to whichever of $n_1 - 1$ and $n_2 - 1$ is smaller.

Test statistic for population proportion difference:

$$\hat{p} = \frac{Y_1 + Y_2}{n_1 + n_2}, \quad Z_{test} = \frac{(\hat{p}_1 - \hat{p}_2) - (\pi_1 - \pi_2)}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$